

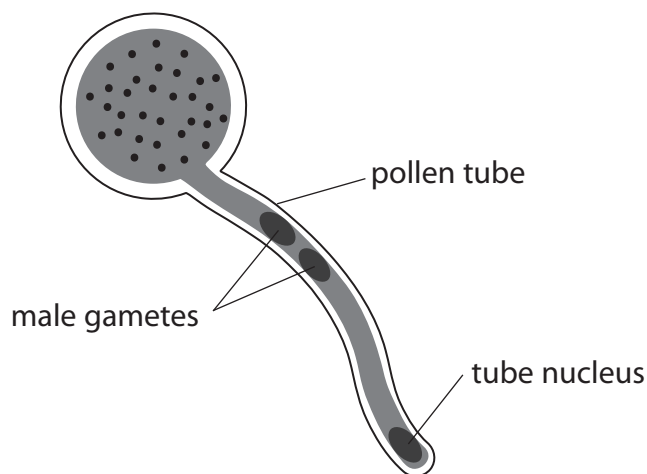
Answer ALL questions.

Write your answers in the spaces provided.

- 1 Pollen grains will germinate, and the pollen tubes will grow, in water but not as quickly as they do on the stigma.

This difference is thought to be due to several factors, such as calcium ions, sucrose and compounds containing boron. These factors are present on the stigma and in the style.

The diagram shows a germinating pollen grain.



- (a) (i) Explain the role of the pollen tube in plant reproduction.

(3)

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- (ii) Explain why pollen tubes grow in the same direction from the stigma, after pollination.

(2)

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- (b) The effect of sucrose solutions of different concentrations on the growth of pollen grains was investigated.

The percentage of pollen grains that germinated and the length of the pollen tubes were measured, one hour after they were placed in the sucrose solution.

- (i) State **two** dependent variables in this investigation.

(2)

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- (ii) Explain why changes in temperature will affect this investigation.

(2)

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(c) A solution containing calcium ions has to be added to the sucrose solutions.

These solutions were made using:

- a 2.0 mol dm^{-3} sucrose solution
- a solution containing calcium ions and water

(i) Describe how you would make 10 cm^3 of a 0.4 mol dm^{-3} sucrose solution using these two solutions.

(3)

(ii) The calcium ion solution is alkaline.

State how the pH of the sucrose solutions used in this investigation could be adjusted to a required value.

(1)



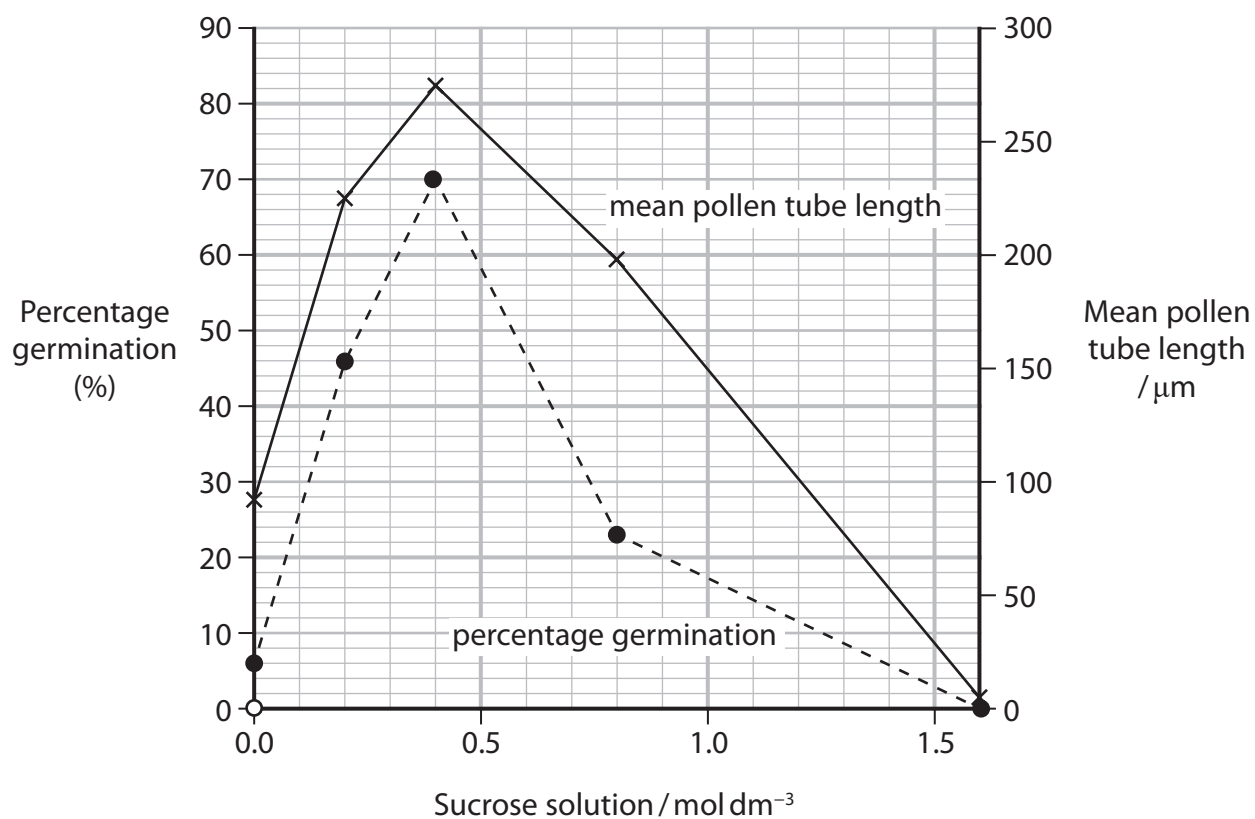
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(d) The graph shows the results of this investigation.



(i) Draw a suitable table to show the results for percentage germination.

(3)



- (ii) Comment on the conclusion that both percentage germination and mean pollen tube growth have the same optimum sucrose concentration.

(3)

(Total for Question 1 = 19 marks)

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